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Customer Segmentation Project Report

Project Overview

Customer segmentation helps businesses understand their customers by grouping them based on shared characteristics. This allows companies to design personalized marketing strategies, improve engagement, and increase revenue.

In this project, we analyze a dataset containing the following features:

- **CustomerID** – Unique identifier for each customer
- **Gender** – Male or Female
- **Profession** – Job category of the customer
- **Spending Score** – Score representing spending habits
- **Age** – Age of the customer

The aim is to **prepare, clean, and explore the dataset** to make it ready for customer segmentation.

1. Data Exploration

- Preview the dataset to check the first few rows
- Examine the data types and count of non-null values
- Review summary statistics for numerical columns, including mean, median, and standard deviation

Key Observations:

- The dataset has X rows and Y columns
- `Gender` and `Profession` are categorical
- `Age` and `Spending Score` are numerical
- No duplicate `CustomerID`s

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2. Handling Missing Data

Missing values can distort analysis, so we handle them carefully:

- Identify columns with missing values
- Replace missing categorical values (like `Profession`) with the most frequent category
- Replace missing numerical values (like `Age`) with mean or median values

Outcome: Dataset has no missing values.

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3. Outlier Detection

- Detect numerical outliers using Z-score and IQR methods
- Use boxplots to visually inspect unusual values

This ensures extreme data points are identified before analysis.

4. Outlier Treatment

After identifying outliers, we handle them appropriately:

- Remove extreme outliers that are likely errors
- Replace moderate outliers with median values
- Use winsorization if necessary to limit extreme effects

This step ensures that data patterns are accurate and reliable.

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5. Feature Scaling

- Normalize features to scale values between 0 and 1
- Standardize features to have a mean of 0 and S.D of 1

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6. Encoding Categorical Variables

Machine learning algorithms require numerical input:

- Convert binary categories (like `Gender`) using label encoding
- Convert multi-category features (like `Profession`) using one-hot encoding

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7. Feature Engineering

Feature engineering enhances the ability to detect meaningful customer segments.

✧ **AgeGroup** – Categorize ages as Young, Middle-aged, or Senior.
✧ **HighSpender** – Identify customers with high Spending Score.

8. Exploratory Data Analysis (EDA)

EDA is used to visualize patterns and trends:

- Plot distributions of `Age` and `Spending Score` to understand their spread.
- Visualize relationships between `Spending Score` and `Age` across different `Gender` and `Profession` categories
- Use boxplots to compare spending patterns across groups

Insights Expected:

1. Certain age groups / professions spend more than others.
2. Spending patterns may suggest natural clusters
3. High-spending customers can be identified for targeted marketing

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9. Final Clean Dataset Preparation

After completing all preprocessing steps:

- Missing values handled
- Outliers treated
- Features scaled and encoded
- New engineered features added

The final dataset is **clean, consistent, and ready** for segmentation algorithms.

10. Conclusion:

This project demonstrates a complete workflow for customer segmentation:

- ✓ **Data Exploration**: Understand the dataset and initial statistics
- ✓ **Data Cleaning**: Handle missing values and outliers
- ✓ **Feature Preparation**: Scale numerical features and encode categorical features
- ✓ **Feature Engineering**: Create new features for better insights
- ✓ **Exploratory Data Analysis**: Visualize trends and patterns

The cleaned and processed dataset is ready for clustering or segmentation, providing high-value customers, understanding spending habits & targeted marketing strategies.